

AADK 5 - Task 1 – Deep Dive: Kotlin Functions, Lambdas & Higher-Order Functions

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Research & Collect Examples

1. Regular Named Function

Definition:

```
fun greet(name: String): String {  
    return "Hello $name"  
}
```

Real-World Use (Android/Compose Context):

Used to format user data before displaying it in a Text() composable.

Example:

```
Text(text = greet("Rajat"))
```

Advantage:

- Clear and readable.
 - Best for reusable logic with descriptive names.
 - Easier to debug compared to inline lambdas.
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2. Lambda Expression

Definition:

```
val greet: (String) -> String = { name -> "Hello $name" }
```

Real-World Use (Compose Example):

Used in button click listeners:

```
Button(onClick = { println("Button Clicked") }) {  
    Text("Click Me")  
}
```

Advantage:

- Concise and inline.
 - Reduces need for anonymous function declarations.
 - Perfect for event handling in UI.
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3. Higher-Order Function

Definition:

```
fun applyOperation(a: Int, b: Int, op: (Int, Int) -> Int): Int {  
    return op(a, b)  
}
```

Usage Example:

```
val result = applyOperation(4, 5) { x, y -> x + y }  
println(result)
```

Real-World Use (Compose Context):

Compose functions like Button() accept lambdas:

```
Button(onClick = { /* action */ }) { }
```

Here, onClick is a lambda parameter — making Button a higher-order function.

Advantage:

- Promotes reusable and flexible code.
 - Enables powerful UI APIs like Compose.
 - Encourages functional programming style.
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Comparison Table

Function Style	Syntax Example	When to Use	Trade-offs
Regular Named Function	<code>fun greet(name: String): String</code>	When logic needs clarity and reuse	Slightly more verbose
Lambda Expression	<code>{ name -> "Hello \$name" }</code>	Short operations, event handlers	Can reduce readability if too complex
Higher-Order Function	<code>fun applyOperation(..., op: (Int, Int) -> Int)</code>	When behavior needs to be passed dynamically	Harder to understand for beginners

Mini-Exercise

Higher-Order Function with Lambda (Compose-style Example) | Extension Function Example

```
Kotlin
2.3.10 JVM Program arguments

fun performAction(action: () -> Unit) {
    action()
}

fun main() {
    performAction {
        println("Action performed!")
    }
}
```

Action performed!

```
Kotlin
2.3.10 JVM Program arguments

fun String.addExclamation(): String {
    return this + "!"
}

fun main() {
    val message = "Hello".addExclamation()
    println(message)
}
```

Hello!

Compose Equivalent Pattern:

```
Button(onClick = {
    println("Button clicked")
}) {
    Text("Click Me")
}
```

Reflection

Lambdas and higher-order functions significantly reduce boilerplate in Compose by eliminating the need for interface implementations like `OnClickListener`. Instead of writing separate classes, we directly pass behavior as parameters. This keeps UI code declarative and compact. Compose heavily relies on this pattern to keep UI definitions clean and readable.

However, excessive nested lambdas (especially in large composables) can reduce readability. When lambdas become too long or contain complex logic, debugging and understanding the flow becomes difficult.
